Psychology 434 Test 2 Study Sheet

Weeks 8–10

1. From ATE to CATE

The average treatment effect is

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)].$$

The conditional average treatment effect is

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

The ATE summarises a population. The CATE summarises sub-populations defined by covariate values. The difference matters when the ATE hides meaningful variation across people.

2. Causal forests

A causal forest learns where in the covariate space $\tau(x)$ varies, by recursively splitting the data on whichever covariates and thresholds reduce within-node heterogeneity in treatment effect. Two design choices matter for inference.

Honest splitting. The tree's structure is learned on one subsample; leaf-level treatment effects are estimated on a different subsample. Honesty prevents the same data from being used to both choose splits and estimate effects, which would bias the estimates and the confidence intervals.

Doubly-robust scores. Each unit's contribution to the estimator is built from both an outcome model and a propensity model. The estimator is consistent if either model is approximately correct. This protects against misspecification of any single component.

3. Heterogeneity as a hypothesis: RATE, AUTOC, Qini

If a forest's predicted $\hat{\tau}(x)$ ranks units by their estimated benefit, the **rank-weighted average treatment effect** (RATE) measures whether the high-ranked units actually have larger treatment effects than the low-ranked units. RATE is an omnibus test of heterogeneity that uses information from across the covariate space.

- **AUTOC** weights by ranks across the whole population (good for detecting heterogeneity).
- **Qini** weights by treated fraction (good for evaluating targeting under a budget constraint).

A small or zero RATE means the forest's ranking is not picking up treatment-effect heterogeneity that is large enough to act on. A wide CI on RATE means the data cannot distinguish heterogeneity from noise.

A **Qini curve** plots cumulative treatment effect against cumulative treated fraction. A curve well above the diagonal indicates targeting helps. A curve that hugs the diagonal indicates targeting and uniform treatment perform similarly.

4. Policy trees

A policy tree converts $\hat{\tau}(x)$ into a deployable allocation rule by partitioning the covariate space into a small number of leaves and assigning each leaf an action (treat / do not treat). It chooses splits that maximise expected policy value, optionally under a budget constraint.

Reading a tree. Each non-leaf node names a covariate and a threshold. Each leaf says "treat" or "do not treat". A depth-2 tree asks at most three yes/no questions before committing.

Choosing depth (parsimony rule). Prefer the simpler depth-1 tree unless the depth-2 tree's policy value is meaningfully larger and its CI excludes the depth-1 estimate. Simpler rules are more interpretable, more stable across resamples, and easier to defend to non-technical decision-makers.

Off-policy evaluation. Policy values are estimated on held-out data not used to construct the rule, so they reflect out-of-sample performance.

5. Equity, governance, democratic judgement

A policy tree maximises expected benefit under its objective. It does not decide what justice requires. Three live threats:

Proxy variables. A split on a deprivation, income, postcode, age, or baseline-wellbeing variable can affect social groups differently even when those groups are not named in the tree. Removing an explicit variable does not remove its proxies.

Most-versus-somewhat trade-off. Targeting those who benefit *most* may withhold treatment from those who benefit *somewhat*. Whether that is acceptable depends on values, law, institutional purpose, and democratic accountability.

Science and public judgement. Statisticians and psychologists can estimate benefits, harms, uncertainty, and subgroup patterns. They do not get to legislate justice. In a democracy, citizens and institutions debate and vote about the values that govern public allocation. The analyst's job is to make the trade-offs visible and to avoid presenting an objective function as a public decision.

A defensible deployment names the decision-maker, the public objective, the override conditions, the audit plan, and the limits of the model.

6. Outcome-wide design and multiple-testing correction

When a single exposure is tested against several outcomes, the chance that at least one CI excludes zero by chance alone exceeds the nominal per-test rate.

Bonferroni correction. With K outcomes at family-wise error rate $\alpha_{FW} = 0.05$, test each outcome at $\alpha = 0.05/K$. For $K = 4$, that is $\alpha = 0.0125$, equivalent to a 98.75% CI per outcome. Report multiplicity-adjusted intervals alongside (or in place of) the unadjusted ones.

The aim of an outcome-wide design is meta-analytic interpretation, not cherry-picking. Read the full pattern of estimates, not the one outcome that survives the cut.

7. Sensitivity to unmeasured confounding: E-values

The **E-value** for an estimated effect is the minimum association strength, on the risk-ratio scale, that an unmeasured confounder would need to have with both the exposure and the outcome — over and above measured covariates — to fully explain away the effect. Larger E-values indicate greater robustness.

Rule-of-thumb interpretation:

- ≥ 2.0: robust to strong unmeasured confounding.
- 1.5–2.0: robust to moderate unmeasured confounding.
- < 1.5: vulnerable to weak unmeasured confounding.

E-values should be reported for the point estimate and for the lower bound of the multiplicity-adjusted CI (the more stringent test).

8. Measurement and causal inference

Reflective vs formative models. A reflective model treats indicators as effects of an underlying latent construct (the construct causes the items). A formative model treats indicators as constituents of the construct (the items together compose it). Each is awkward for causal inference: reflective models posit a single latent cause that may not exist as a real-world quantity; formative models lack a clear potential-outcomes interpretation, because intervening on a construct is not the same as intervening on each item.

Multiple versions of treatment. When one treatment label bundles substantively different versions, $Y(1)$ is not well-defined. Consistency is threatened. The recorded ATE is then a population-weighted average across versions rather than the effect of any single intervention.

Measurement invariance. Invariance asks whether items relate to the construct the same way across groups. Configural invariance fails if items load on different latent structures; metric invariance fails if loadings differ; scalar invariance fails if intercepts differ. Without scalar invariance, mean differences across countries or languages confound true differences with measurement differences.

Measurement error as a structural threat. Let Y^* be the true outcome and Y the recorded outcome. If exposure A also affects how Y is recorded (e.g., the intervention teaches participants to label distress differently), the path $A \to U_Y \to Y$ is causal but does not run through Y^*. Identification of the effect on Y^* then requires extra assumptions or design responses.

Confounding control by baseline measures. In a three-wave panel, including the baseline measures of both the exposure (A_0) and the outcomes ($Y_{k,0}$) in the adjustment set provides strong confounding control. Any unmeasured confounder would need to act on later exposure conditional on baseline exposure and baseline outcome — a demanding requirement.

9. How to study

For each scenario you encounter:

1. State the causal question and the causal estimand. Distinguish the causal estimand from the statistical one.
2. Draw the DAG. Name the adjustment set the DAG implies.
3. State the identification assumptions: consistency, exchangeability, positivity. Add the design responses (lagged-self adjustment, IPCW for attrition).
4. Choose an estimator. For heterogeneity, name the omnibus test (RATE) and the targeting tool (policy tree).
5. Apply the parsimony rule for depth, the Bonferroni correction for multiple outcomes, and the E-value for sensitivity.
6. Conduct an equity audit before recommending deployment. State one value judgement the model cannot settle.

7. Comment on measurement: which version of the treatment, which scale, which invariance level holds.

Practise with new examples rather than only rereading definitions. For any scenario, ask what could go wrong at each step, and what evidence would change your answer.